Voltage Unbalance Mitigation in LV Networks Using Three-Phase PV Systems

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***Abstract*— In this paper a new method is proposed to mitigate voltage unbalance caused by single-phase solar inverters in low voltage (LV) networks. The method is based on uneven reactive power absorption and injection by three-phase solar inverters. Independent control of each phase is performed to achieve this uneven injection. The average values of phase voltages at the connection points of the photovoltaic (PV) inverters are used as the references for the balancing algorithm. Voltage unbalance mitigation is achieved by use of this method in different scenarios with variable three-phase and single-phase inverters penetration in a realistic LV grid. In addition, the overvoltage is reduced by using this method.**

Keywords*— *voltage unbalance, LV grids, photovoltaic, reactive power, voltage control.

I. INTRODUCTION

Solar energy has surged in use over the past decade. Global PV capacity has evolved from 23 GW in 2009 to 138.9 GW in 2013, and the forecast for PV installation in the next 5 years indicates a growing trend. PV installations connected to the LV network, such as rooftop solar energy, comprises the majority of the European market, and show a growing trend [1]. These PV installations in the LV network can be either connected as three-phase or single-phase systems. Due to a growth in single-phase PV systems penetration, voltage unbalance ratio can also experience an increment in LV networks, because of the different active power flow in each phase [2].

A three-phase system undergoes voltage unbalance when the magnitudes of phase or line voltages are different, when the phase angles differ from the balanced conditions, or when they both occur at the same time [3]. Traditionally, voltage unbalance existed under unbalanced load scenarios, where power demand was not equal among the three lines. Under single-phase PV systems power injection, three-phase distribution systems may also experience uneven active power flow, contributing to voltage unbalance.

In unbalance situations overheating and losses increase can be observed in three-phase motors present on LV networks [4]. Single-phase loads, such as electronic devices, can also be affected by three-phase voltage unbalance if the phase they are connected to experiences overvoltage or undervoltage [5]. Together with PV systems penetration increase in LV networks, control systems can be deployed to mitigate inconveniences associated with this increase.

Previous research has been undertaken regarding voltage unbalance mitigation. In [7] DSTATCOM technology is applied to solve voltage unbalance caused by single-phase PV inverters in LV networks. The main idea in this method is to maintain the voltage level in the connection point close to 1 p.u., injecting both active and reactive power. Nevertheless, DSTATCOM devices placement in relation with the voltage unbalance sources is a key factor to its efficiency, and the related cost is high. In [8] single-phase batteries are used as energy storage systems to mitigate voltage unbalance. Reducing the neutral current to zero is the cornerstone of this method. The inverter will control active and reactive power injection to achieve this target. However, its reliability is related to the location of the storage system in the network. On the other hand, current measurements in the neutral wire are not always available. Additionally, the investment needed to install the batteries is high. Finally, in [9] electric vehicles (EV) are used to solve voltage unbalance. In this method, EVs charger and batteries inject and absorb both reactive and active power, instead of managing only active power flow. The cornerstone of the method is the control on reactive power flow in order to compensate the uneven active power flow derived from the arbitrary demand in each phase. This idea could be implemented beyond the idea of mitigating voltage unbalance caused by EVs. However, its deployment is closely related to EVs penetration in modern societies.

The inherent scalability of solar inverters, the possibility of installing them among the LV distribution network and their capability of controlling reactive power independently from active power generation confers PV systems an advantageous position to act as voltage unbalance controllers. The fact that they can be placed evenly distributed among the LV network ensures a better respond than the other methods already studied to the need of the control system to be close to the voltage unbalance source. On the other hand, the estimated cost derived of using PV inverters as controllers only depends on oversizing the inverter, which a priori seems less expensive than installing new devices. Finally, PV inverters, differently from EVs, have already deployed, ensuring the possibility of using these devices as network operators in the short term.

In this paper, a new method is proposed to solve voltage unbalance caused by single-phase inverters penetration by means of reactive power control in three-phase PV inverters. The key idea of the deployed method is uneven absorption or injection of reactive power in each of the three phases and

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independent control of each phase will be considered. The average value of the three phases' voltage magnitude will be used as a reference for balancing the voltages in the connection point. The method relies on three-phase inverters distributed within the LV network. This restricts the deployment of the method, but in return the method is relatively cost-effective because oversizing inverters and adding additional control is the only associated cost.

The structure of this paper is as follows. First, a debate about voltage unbalance definitions is presented in chapter II. An explanation of the deployed method follows in chapter III. Chapter IV provides information about the LV network used for simulation, as well as the results of the simulations performed in this network. Chapter V includes a conclusion and future work overview.

II. VOLTAGE UNBALANCE DEFINITIONS

Three different voltage unbalance definitions are given by three different communities [3]. Firstly, the national equipment manufacturers' association (NEMA) provides a definition regarding line magnitude voltages (LVUR).

$$\mathrm{LVUR}(\%) = \frac{\text{max voltage deviation from the avg line voltage}}{\text{avg line voltage}} \times 100 \quad (1)$$

Secondly, IEEE's definition has regard to phase magnitude voltage (PVUR)

$$\mathrm{PVUR}(\%) = \frac{\text{max voltage deviation from the avg phase voltage}}{\text{avg phase voltage}} \times 100 \quad (2)$$

Finally, the power community provides a definition involving positive and negative sequences.

$$\mathrm{VUF}(\%) = \frac{\text{negative sequence voltage component}}{\text{positive sequence voltage component}} \times 100 \quad (3)$$

As it can be seen, voltage unbalance definitions given by these different communities are not consistent and there is no unanimous assent in the scientific community about voltage unbalance definition. Several studies have been undertaken in order to analyze the relation between these definitions [10], [11]. Moreover, in [12] new unbalance definitions are proposed. In [13] a comparison of the different definitions' behavior under different unbalance cases is given. A case study is done in [13] analyzing overvoltage magnitude unbalance, which due to the mainly resistive nature of the LV networks, is close to the type of unbalance caused by single-phase PV systems in LV networks. For this scenario, PVUR is the definition which provides a highest unbalance ratio. Thus, the method proposed in this paper will use the PVUR unbalance definition.

III. PROPOSED METHOD

Regulation of voltage level has been proven to be possible by means of reactive power control in three-phase balanced networks [14]. Several studies have been undertaken regarding overvoltage control under PV penetration by use of reactive power [14]-[18]. In [14], voltage level is maintained within the permitted levels by injecting reactive power when the voltage decreases and absorbing reactive power when voltage level is over the rated value. The basis of the idea deployed in this paper is to extrapolate the interaction between voltage level and reactive power flow in a three-phase balanced network to each phase independently. The cornerstone of the balancing method is to equilibrate the three-phase voltages around the average value of the three of them. For this purpose, reactive power will be injected in those phases whose voltage magnitude is under the average value, whereas reactive power absorption will be performed in those phases where the voltage levels are higher than the average. Independent control of each phase will be accomplished, as it can be seen in Fig. 1. In this figure, Ua, Ub and Uc represent the phase voltages, Uavg is the average value of the three voltages, and Qcmd a, Qcmd b and Qcmd c are the calculated reactive powers that have to be injected into each phase.

Measurements of voltage magnitude in each phase are done by three-phase inverters. PVUR ratio is calculated, activating the control system when PVUR value surpasses 2% legal limit established by standard EN 50160 [19]. Voltage magnitudes are then compared in each phase with the average value of the three- phase voltages. Reactive power injection or absorption is done in each phase depending on the voltage/average voltage ratio for each of them according to a function shown in Fig. 2.

The average value of the three-phase voltages are used as a reference to balance the voltage. On the one hand, this contributes to minimize reactive power injection and absorption in each situation, as it is not a fixed value and it varies for each situation. On the other hand, it ensures the steady state value remains within the voltage limits. When operating under overvoltage circumstances, the inverters are disconnected, turning useless to study voltage unbalance mitigation in these situations. For this reason, the average voltage value will always remain within the limit when voltage unbalance needs to be solved.

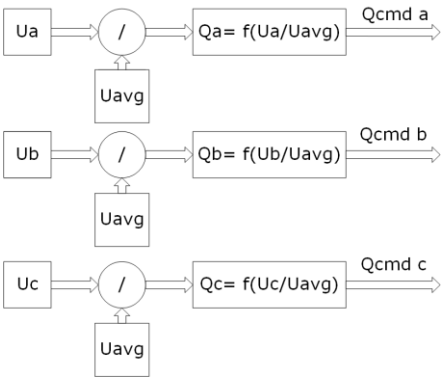


Fig. 1 Scheme of the control system used in this method

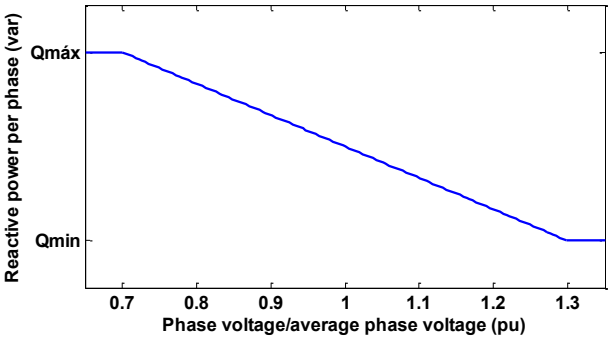


Fig. 2 Basic function relating reactive power and voltage level.

IV. Simulation Results

A. The test system

A branch of Bornholm distribution network is replicated for simulations. Bornholm is a Danish island which represents 1% of the total of the Danish population, area and electricity demand. Together with this, Bornholm's distribution network gathers the main characteristics of the Danish distribution network, turning Bornholm into a useful platform to test the method's respond in a real scenario [20].

The chosen branch is depicted in Fig. 3. It corresponds to a 71-consumers residential area. Two feeders supply energy to the consumers in radial configuration. Appendix 1 contains cables parameters. Load consumption is neglected to enable an unconditioned study of the impact of the controller.

PV generation data is captured from a real PV rooftop system placed at DTU campus in Lyngby. These data correspond to measurements performed every 15 minutes in a cloudless day and is shown in Fig. 4. 5kW PV installation per consumer is considered as 100% PV penetration in this network. The definition of PV penetration in this paper is as follows:

$$\text{PV penetration}\,(\%) = \frac{P_{\text{installed}}}{5\text{kW}} \times 100 \quad (4)$$

Where $P_{installed}$ corresponds to the total PV power installed per household. 30% and 40% PV penetration are contemplated in the simulations. Both single-phase and three-phase installations are considered. As there are several distribution possibilities when considering single-phase and three-phase PV systems, the same penetration of both of them has been selected for all consumers. The penetration of single-phase inverters is defined in the formula below and it varies in every case studies. All single-phase inverters are connected in phase A in order to replicate the worst case scenario for voltage unbalance. In addition, it is supposed that the single-phase inverters do not inject or absorb reactive power and work at unity power factor.

$$\text{Single - phase penetration}(\%) = \frac{P_{\text{single - phase}}}{P_{\text{total}}} \times 100 \quad (5)$$

Where P_{total} corresponds to the global PV power installed per household, considering both single-phase and three-phase systems. Three representative branches of the test system, highlighted in Fig. 3, are evaluated in each study case with the intention of assessing voltage unbalance evolution among the network. In addition, parameters in end of feeder nodes are evaluated, as voltage unbalance attains its highest level in this location. Fig. 5 proves this assessment by showing voltage unbalance ratio at the time of the day when its highest amount is reached among the three branches indicated in Fig. 3.

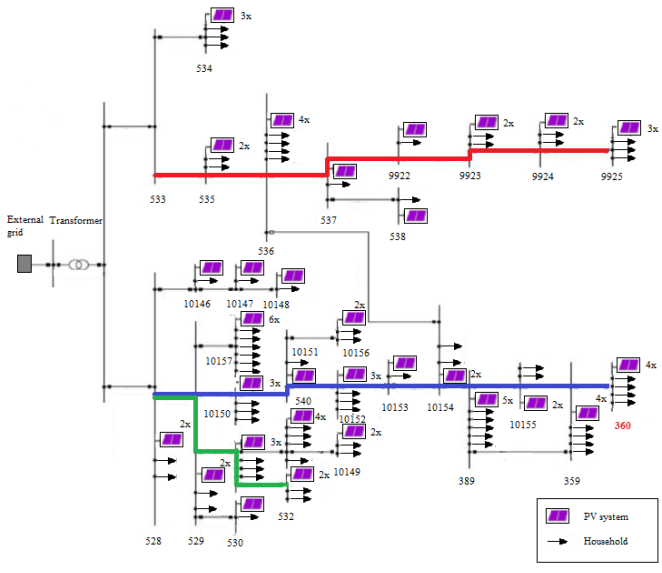


Fig. 3 Bornholm Low Voltage Network under study

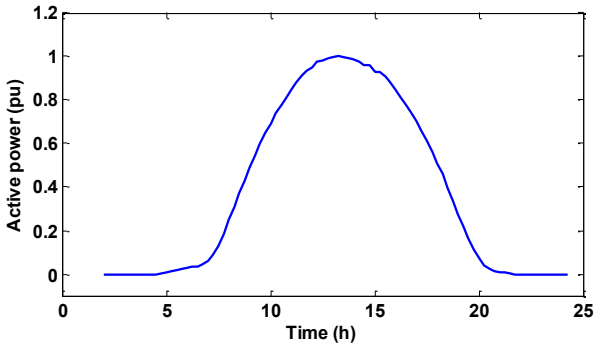


Fig. 4 Normalized injection curve for a PV inverter in a cloudless day.

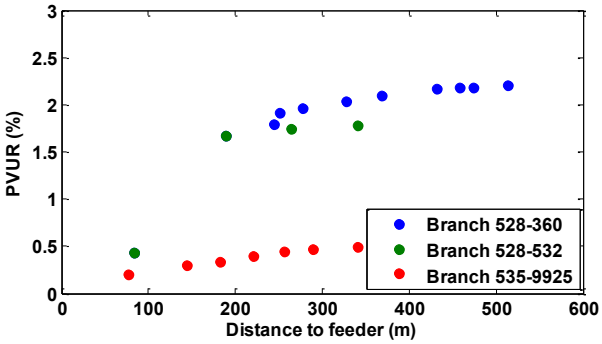


Fig. 5 PVUR ratio evolution among three representative branches.

B. Simulation case studies

In order to improve the efficiency of the method, an enhanced version of the function shown in Fig. 2 is used for the simulation which is depicted in Fig. 6. The reactive power limits shown in Fig. 6 are chosen to comply with the requisites included in [21].

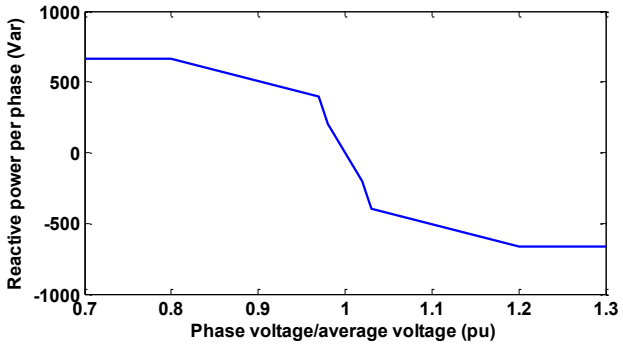


Fig. 6 Reactive power injection curve Q=f(U/Uavg) used for simulations.

In table I, the results of different simulations are shown. For each simulation case, single-phase penetration varies. Two different PV penetrations are considered for these simulations. The results show the highest voltage unbalance ratio reached during the day with and without control applied in node 360 for each case. This is the node which experiences the highest unbalance ratio in the test network, as it can be seen in Fig. 5.

Voltage unbalance decrease is observed in all simulation cases after the control system is applied. Nevertheless, for two case studies the PVUR ratio after applying the control is higher than the 2% legal limit. Variations on the function slope can be considered in order to increase the reactive power managed and improve the results. Parameters for case study 3 (40% PV-10% single-phase) are depicted in Fig. 7-10.

By comparing Fig. 7 and Fig. 8, it can be observed that voltage unbalance ratio is decreased among the branches, remaining below 2% limit in every node of the studied branches.

As it can be seen in Fig. 7 and Fig. 8, voltage unbalance ratio is decreased also in those nodes of the branches which did not present PVUR ratio over the legal limit. In those nodes there is no reactive power injection and absorption by the inverters, but it can be concluded that the control system activated in other nodes in this network has a positive effect on the voltage unbalance ratio for the rest of the nodes in the branches.

TABLE I SIMULATION RESULTS

PV (%)	Single-Phase (%)	PVUR no-control (%)	PVUR with control (%)
30%	10%	2.21	1.87
	17%	3.72	2.08
40%	10%	2.75	1.98
	15%	4.15	2.26

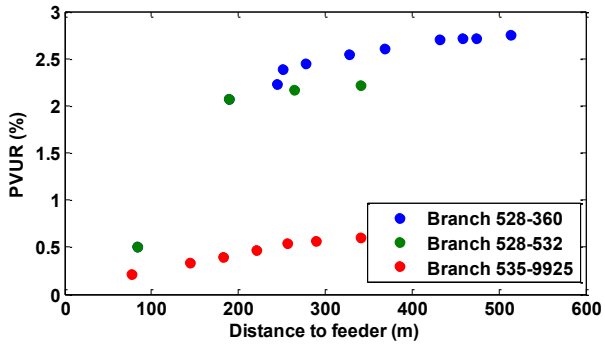


Fig. 7 PVUR evolution across the branches without control.

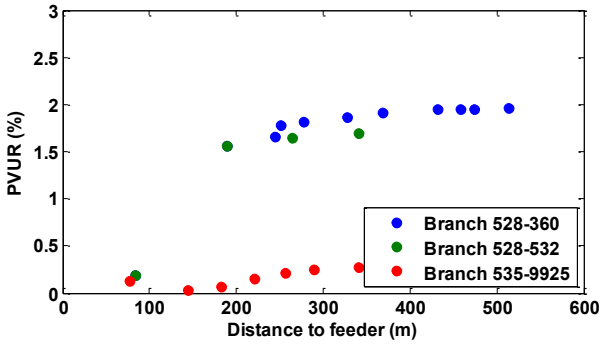


Fig. 8 PVUR evolution across the branches with control.

Fig. 9 shows voltage evolution during 24 hours in node 360. The voltage evolution without applying the control system is shown in dashed lines. Continuous lines are used to show the voltage level in the controlled network. As a result of the uneven reactive power injection, the voltage in phase A decreases, while the voltage in phase C increases and the voltage in phase B maintains its value. It can be seen in Fig. 9 that voltage in phase A suffers overvoltage over the 253 V permitted limit, which is marked in a purple line in the figure. The single-phase overvoltage in phase A occurs due to single-phase inverters connected to this phase. After the control is applied, the voltage level in phase A decreases from 255.40 V to 253.04 V, remaining below that limit, as it is shown in Fig. 9. It can be concluded that this method can solve single-phase overvoltage at the same time as solving voltage unbalance.

Fig. 10 shows the reactive power injection and absorption per phase for this case study in node 360. This corresponds to 4 inverters' injection. It can be appreciated in Fig. 10 the independent control of each phase. Phase A, whose voltage

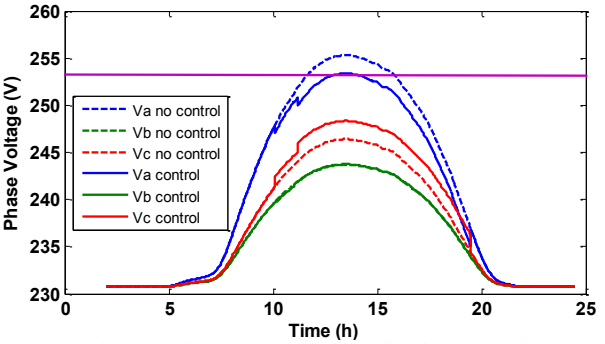


Fig. 9 Voltage profile in node 360 with and without control.

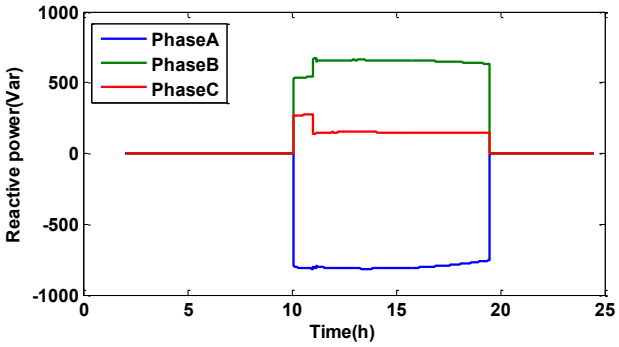


Fig. 10 Reactive power injection per phase in node 360.

value is over the average, absorbs reactive power while phases B and C inject it, since their voltage values are lower than the average. The maximum reactive power injected for this case study in node 360 is 659.5, which corresponds to 165 Var per household. The maximum reactive power absorbed is 813 Var, approximately 203 Var per inverter household.

V. Conclusion

In this paper a method was proposed to solve voltage unbalance in LV networks due to single-phase solar inverters penetration. Single-phase PV systems can worsen the voltage unbalance ratio, increasing it over 2% legal limit. By use of the method proposed in this paper, voltage unbalance was reduced in all cases. Voltage unbalance ratio was remained under 2% permitted limit in some of the study cases, allowing single-phase solar inverters penetration to increase. Moreover, single-phase overvoltage mitigation generated by single-phase solar inverters penetration is also achieved by use of this method. Combination of uneven reactive power injection with uneven active power injection and distributing the active power between the phases when the solar generation allows can also be studied to pursue a more efficient voltage unbalance mitigation method. Aside from possible improvements on the method, studies regarding the effect of reactive power injection on the medium voltage network can also be done.

Appendix

TABLE II Cable Parameters

Cable	Type	Length (km)	Resistance (Ohm/km)	Reactance (Ohm/km)
125LV_528	4X150 AL-M PEX	0.084	0.207	0.078
125LV_533	4X95 AL-M PVICS	0.03	0.321	0.082
528_529	4X50 AL-M PEX	0.105	0.641	0.081
528_1046	4X50 AL-M PEX	0.033	0.641	0.081
533_535	4X95 AL-M PVICS	0.047	0.321	0.082
533_534	4X50 AL-M PVICS	0.103	0.641	0.084
529_530	4X50 AL-M PEX	0.053	0.641	0.081
529_531	4X95 AL-M PVICS	0.076	0.321	0.082
529_10150	4X150 AL-M PEX	0.055	0.207	0.078
529_10157	4X50 AL-M PEX	0.003	0.207	0.078
10146_10147	4X50 AL-M PEX	0.068	0.641	0.081
535_536	4X95 AL-M PVICS	0.068	0.321	0.082
531_532	4X50 AL-M PEX	0.076	0.641	0.081
531_540	4X50 AL-M PEX	0.056	0.641	0.081
10150_10151	4X150 AL-M PEX	0.07	0.207	0.078
10147_10148	4X50 AL-M PEX	0.072	0.641	0.081
540_10149	4X150 AL-M PEX	0.032	0.207	0.078
10151_10152	4X150 AL-M PEX	0.026	0.207	0.078
10151_10156	4X50 AL-M PEX	0.062	0.641	0.081
536_537	4X95 AL-M PVICS	0.038	0.321	0.082
10152_10153	4X150 AL-M PEX	0.051	0.207	0.078
537_538	4X50 AL-M PVICS	0.065	0.641	0.084
537_9922	4X50 AL-M PVICS	0.038	0.641	0.084
10153_10154	4X150 AL-M PEX	0.041	0.207	0.078
9922_9923	4X50 AL-M PVICS	0.035	0.641	0.084
10154_389	4X150 AL-M PEX	0.063	0.207	0.078
389_359	4X150 AL-M PVICS	0.042	0.207	0.084
389_10155	4X150 AL-M PEX	0.026	0.207	0.078
10155_359	4X150 AL-M PEX	0.02	0.207	0.078
9923_9924	4X95 AL-M PEX	0.033	0.321	0.078
359_360	4X50 AL-M PVICS	0.035	0.641	0.084
9924_9925	4X95 AL-M PEX	0.052	0.321	0.078

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